G. H. Hardy (2017). *A Mathematician's Apology*

Collector: Nguyen Quan Ba Hong

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Foreword

"It was a perfectly ordinary night at Christ's high table, except that Hardy was dining as a guest.

He had just returned to Cambridge as Sadleirian professor, and I had heard something of him from young Cambridge mathematicians.

They were delighted to have him back: he was a *real* mathematician, they said, not like those Diracs and Bohrs the physicists were always talking about: *he was the purest of the pure.*

He was also unorthodox, eccentric, radical, ready to talk about anything.

This was 1931, and the phrase was not yet in English use, but in later days they would have said that in some indefinable way he had star quality.

So, from lower down the table, I kept studying him.

He was then in his early 50s: his hair was already grey, above skin so deeply sunburnt that it stayed a kind of Red Indian bronze.

His face was beautiful - high cheek bones, thin nose, spiritual and austere but capable of dissolving into convulsions of internal gamin-like amusement.

He had opaque brown eyes, bright as a bird's - a kind of eye not uncommon among those with a gift for conceptual thought.

Cambridge at that time was full of unusual and distinguished faces - but even then, I thought that night, Hardy's stood out.

I do not remember what he was wearing.

It may easily have been a sports coat and grey flannels under his gown.

Like Einstein, he dressed to please himself: though, unlike Einstein, he diversified his casual clothing by a taste for expensive silk shirts.

As we sat round the combination-room table, drinking wine after dinner, someone said that Hardy wanted to talk to me about cricket.

I had been elected only a year before, but Christ's was then a small college, and the pastimes of even the junior fellows were soon identified.

I was taken to sit by him.

I was not introduced.

He was, as I later discovered, shy and self-conscious in all formal actions, and had a dread of introductions.

He just put his head down as it were in a butt of acknowledgment, and without any preamble whatever began:

> 'You're supposed to know something about cricket, aren't you?' Yes, I said, I knew a bit.

Immediately he began to put me through a moderately stiff viva.

Did I play?

What sort of performer was I?

I half-guessed that he had a horror of persons, then prevalent in academic society, who devotedly studied the literature but had never played the game.

I trotted out my credentials, such as they were.

He appeared to find the reply partially reassuring, and went on to more tactical questions.

Whom should I have chosen as captain for the last test match a year before (in 1930)?

If the selectors had decided that Snow was the man to save England, what would have been my strategy and tactics?

('You are allowed to act, if you are sufficiently modest, as non-playing captain.')

And so on, oblivious to the rest of the table.

He was quite absorbed.

As I had plenty of opportunities to realize in the future, Hardy had no faith in intuitions or impressions, his own or anyone else's.

The only way to assess someone's knowledge, in Hardy's view, was to examine him.

That went for mathematics, literature, philosophy, politics, anything you like.

If the man had bluffed and then wilted under the questions, that was his lookout.

1st things came 1st, in that brilliant and concentrated mind.

That night in the combination-room, it was necessary to discover whether I should be tolerable as a cricket companion.

Nothing else mattered.

In the end he smiled with immense charm, with child-like openness, and said that Fenner's (the university cricket ground) next season might be bearable after all, with the prospect of some reasonable conversation.

Thus, just Is I owed my acquaintanceship with Lloyd George to his passion for phrenology, I owed my friendship with Hardy to having wasted a disproportionate amount of my youth on cricket.

I don't know what the moral is.

But it was a major piece of luck for me.

This was intellectually the most valuable friendship of my life.

His mind, as I have just mentioned, was brilliant and concentrated: so much so that by his side anyone else's seemed a little muddy, a little pedestrian and confused.

He wasn't a great genius, as Einstein and Rutherford were.

He said, with his usual clarity, that if the word meant anything he was not a genius at all.

At his best, he said, he was for a short time the 5th best pure mathematician in the world.

Since his character was as beautiful and candid as his mind, he always made the point that his friend and collaborator Littlewood was an appreciably more powerful mathematician than he was, and that his protégé Ramanujan really had natural genius in the sense (though not to the extent, and nothing like so effectively) that the greatest mathematicians had it.

People sometimes thought he was under-rating himself, when he spoke of these friends.

It is true that he was magnanimous, as far from envy as a man can be: but I think one mistakes his quality if one doesn't accept his judgment.

I prefer to believe in his own statement in *A Mathematician Apology*, at the same time so proud and so humble:

> 'I still say to myself when I am depressed and find myself forced to listen to pompous and tiresome people, "Well, I have done 1 thing you could never have done, and that is to have collaborated with Littlewood and Ramanujan on something like equal terms."

In any case, his precise ranking must be left to the historians of mathematics (though it will be an almost impossible job, since so much of his best work was done in collaboration).

There is something else, though, at which he was clearly superior to Einstein or Rutherford or any other great genius: and that is at *turning any work of the intellect, major or minor or sheer play, into a work of art.*

It was that gift above all, I think, which made him, almost without realizing it, purvey such intellectual delight.

When *A Mathematician's Apology* was 1st published, Graham Greene in a review wrote that long with Henry James's notebooks, this was the best account of what it was like to be a *creative artist.*

Thinking about the effect Hardy had on all those around him, I believe that is the clue.

He was born, in 1877, into a modest professional family.

His father was Bursar and Art Master at Cranleigh, then a minor public (English for private) school.

His mother had been senior mistress at the Lincoln Training College for teachers.

Both were gifted and mathematically inclined.

In his case, as in that of most mathematicians, the gene pool doesn't need searching for.

Much of his childhood, unlike Einstein's, was typical of a future mathematician's.

He was demonstrating a formidably high I.Q. as soon as, or before, he learned to talk.

At the age of 2 he was writing down numbers up to millions (a common sign of mathematical ability).

When he was taken to church he amused himself by factorizing the numbers of the hymns: he played with numbers from that time on, a habit which led to the touching scene at Ramanujan's sick-bed: the scene is well known, but later on I shall not be able to resist repeating it.

It was an enlightened, cultivated, highly literate Victorian childhood.

His parents were probably a little obsessive, but also very kind.

Childhood in such a Victorian family was as gentle a time as anything we could provide, though probably intellectually somewhat more exacting.

His was unusual in just 2 respects.

In the 1st place, he suffered from an acute self-consciousness at an unusually early age, long before he was 12.

His parents knew he was prodigiously clear, and so did he.

He came top of his class in all subjects.

But, as the result of coming top of his class, he had to go in front of the school to receive prizes: and that he could not bear.

Dining with me 1 night, he said that he deliberately used to try to get his answers wrong so as to be spared this intolerable ordeal.

His capacity for dissimulation, though, was always minimal: he got the prizes all the same.

Some of this self-consciousness wore off.

He became competitive.

As he says in the *Apology*:

> 'I do not remember having felt, as a boy, any *passion* for mathematics, and such notions as I may have had of the career of a mathematician were far from noble.
>
> I thought of mathematics in terms of examinations and scholarships: I wanted to beat other boys, and this seemed to be the way in which I could do so most decisively.'

Nevertheless, he had to live with an over-delicate nature.

He seems to have been born with 3 skins too few.

Unlike Einstein, who had to subjugate his powerful ego in the study of the external world before he could attain his moral stature, Hardy had to strengthen an ego which wasn't much protected.

This at times in later life made him self-assertive (as Einstein never was) when he had to take a moral stand.

On the other hand, it gave him his introspective insight and beautiful candor, so that he could speak of himself with absolute simplicity (as Einstein never could).

I believe this contradiction, or tension, in this temperament was linked with a curious tic in his behavior.

He was the classical anti-narcissist.

He could not endure having his photograph taken: so far as I know, there are only 5 snapshots in existence.

He would not have any looking glass in his rooms, not even a shaving mirror.

When he went to a hotel, his 1st action was to cover all the looking-glass with towels.

This would have been odd enough, if his face had been like a gargoyle: superficially it might seem odder, since all his life he was good-looking quite out of the ordinary.

But, of course, narcissism and anti-narcissism have nothing to do with looks as outside observers see them.

This behavior seems eccentric, and indeed it was.

Between him and Einstein, though, there was a difference in kind.

Those who spent much time with Einstein - e.g. Infeld - found him grow stranger, less like themselves, the longer they knew him.

I am certain that I should have felt the same.

With Hardy the opposite was true.

His behavior was often different, bizarrely so, from ours: but it came to seem a kind of superstructure set upon a nature which wasn't all that different from our own, except that it was more delicate, less padded, finer-nerved.

The other unusual feature of his childhood was more mundane: but it meant the removal of all practical obstacles throughout his entire career.

Hardy, with his limpid honesty, would have been the last man to be finicky on this matter.

He knew what privilege meant, and he knew that he had possessed it.

His family had no money, only a schoolmaster's income, but they were in touch with the best educational advice of late 19th-century England.

That particular kind of information has always been more significant in this country than any amount of wealth.

The scholarships have been there all right, if one knew how to win them.

There was never the slightest chance of the young Wells or the young Einstein.

From the age of 12 he had only to survive, and his talents would be looked after.

At 12, in fact, he was given a scholarship at Winchester, then and for long afterwards the best mathematical school in England, simply on the strength of some mathematical work he had done at Granleigh.

(Incidentally, one wonders if any great school could be so elastic nowadays?)

There he was taught mathematics in a class of one: in classics he was as good as the other top collegers.

Later, he admitted that he had been well-educated, but he admitted it reluctantly.

He disliked the school, except for its classes.

Like all Victorian public schools, Winchester was a pretty rough place.

He nearly died 1 winter.

He envied Littlewood in his cared-for home as a day boy at St Paul's or other friends at our free-and-easy grammar schools.

He never went near Winchester after he had left it: but he left it, with the inevitability of one who had got on to the right tramlines, with an open scholarship to Trinity.

He had 1 curious grievance against Winchester.

He was a natural ball-games player with a splendid eye.

In this 50s he could usually beat the university 2nd string at real tennis, and in his 60s I saw him bring off startling shots in the cricket nets.

Yet he had not had an hour's coaching at Winchester: his method was defective: if he had been coached, he thought, he would have been a really good batsman, not quite 1st-class, but not too far away.

Like all his judgments on himself, I believe that one is quite true.

It is strange that, at the zenith of Victorian games-worship, such a talent was utterly missed.

I supposed no one thought it worth looking for in the school's top scholar, so frail and sickly, so defensively shy.

It would have been natural for a Wykehamist of his period to go to New College.

That wouldn't have made much difference to his professional career (though, since he always liked Oxford better than Cambridge, he might have stayed there all his life, and some of us would have missed a treat).

He decided to go to Trinity instead, for a reason that he describes, humorously but with his usual undecorated truth, in the *Apology*.

> 'I was about 15 when (in a rather odd way) my ambitions took a sharper turn.
>
> These is a book by "Alan St Aubyn" (actually Mrs Frances Marshall) called *A Fellow of Trinity*, 1 of a series dealing with what is supposed to be Cambridge college life...
>
> There are 2 heroes, a primary hero called Flowers, who is almost wholly good, and a secondary hero, a much weaker vessel, called Brown.
>
> Flowers and Brown find many dangers in university life...
>
> Flowers survives all these troubles, is Second Wrangler and succeeds automatically to a Fellowship (as I suppose he would have done then).
>
> Brown succumbs, ruins his parents, takes to drink, is saved from delirium tremens during a thunderstorm only by the prayers of the Junior Dean, has much difficulty in obtaining even an Ordinary Degree, and ultimately becomes a missionary.
>
> The friendship is not shattered by these unhappy events, and Flowers's thoughts stray to Brown, with affectionate pity, as he drinks port and eats walnuts for the 1st time in Senior Combination Room.
>
> 'Now Flowers was a decent enough fellow (so far as "Alan St Aubyn" could draw one), but even my unsophisticated mind refused to accept him as clever.
>
> If he could do these things, why not I?
>
> In particular, the final scene in Combination Room fascinated me completely, and from that time, until I obtained one, mathematics meant to me primarily a Fellowship of Trinity.'

Which he duly obtained, after getting the highest place in the Mathematical Tripos Part II, at the age of 22.

On the way, there were 2 minor vicissitudes.

The 1st was theological, in the high Victorian manner.

Hardy had decided - I think before he left Winchester - that he did not believe in God.

With him, this was a black-and-white decision, as sharp and clear as all other concepts in his mind.

Chapel at Trinity was compulsory.

Hardy told the Dean, no doubt with his own kind of shy certainty, that he could not conscientiously attend.

The Dean, who must have been a jack-in-office, insisted that Hardy should write to his parents and tell them so.

They were orthodox religious people, and the Dean knew, and Hardy knew much more, that the news would give them pain - pain such as we, 70 years later, cannot easily imagine.

Hardy struggled with his conscience.

He wasn't worldly enough to slip the issue.

He wasn't even worldly enough - he told me 1 afternoon at Fenner's, for the wound still rankled - to take the advice of more sophisticated friends, e.g. George Trevelyan and Desmond MacCarthy, who would have known how to handle the matter.

In the end he wrote the letter.

Partly because of that incident, his religious disbelief remained open and active ever after.

He refused to go into any college chapel even for formal business, like electing a master.

He had clerical friends, but God was his personal enemy.

In all this there was an echo of the 19th century: but one would be wrong, as always with Hardy, not to take him at his word.

Still, he turned it into high jinks.

I remember, 1 day in the 30s, seeing him enjoy a minor triumph.

It happened in a Gentleman v.

Players match at Lord's.

It was early in the morning's play, and the sun was shining over the pavilion.

1 of the batsmen, facing the Nursery end, complained that he was unsighted by a reflection from somewhere unknown.

The umpires, puzzled, padded round by the sight-screen.

Motor-cars?

No.

Windows?

None on that side of the ground.

At last, with justifiable triumph, an umpire traced the reflection down - it came from a large pectoral cross reposing on the middle of an enormous clergyman.

Politely the umpire asked him to take it off.

Close by, Hardy was doubled up in Mephistophelian delight.

That lunch time, he had no leisure for eating: he was writing postcards (postcards and telegrams were his favorite means of communication) to each of his clerical friends.

But in his war against God and God's surrogates, victory was not all on 1 side.

On a quiet and lovely May evening at Fenner's, round about the same period, the chimes of 6 o'clock fell across the ground.

'It's rather unfortunate,' said Hardy simply, 'that some of the happiest hours of my life should have been spent within sound of a Roman Catholic church.'

The 2nd minor upset of his undergraduate years was professional.

Almost since the time of Newton, and all through the 19th century, Cambridge had been dominated by the examination for the old Mathematical Tripos.

The English have always had more faith in competitive examinations than any other people (except perhaps in Imperical Chinese): they have conducted these examinations with traditional justice: but they have often shown remarkable woodenness in deciding what the examinations should be like.

I.e., incidentally, true to this day.

It was certainly true of the Mathematical Tripos in its glory.

It was an examination in which the questions were usually of considerable mechanical difficulty - but unfortunately did not give any opportunity for the candidate to show mathematical imagination or insight or any quality that a creative mathematicians needs.

The top candidates (the Wranglers - a term which still survives, meaning a 1st Class) were arranged, on the basis of marks, in strict numerical order.

Colleges had celebrations when 1 of their number became Senior Wrangler: the 1st 2 or 3 Wranglers were immediately elected Fellows.

It was all very English.

It had only 1 disadvantage, as Hardy pointed out with his polemic clarity, as soon as he had become an eminent mathematician and was engaged, together with his tough ally Littlewood, in getting the system abolished: it had effectively ruined serious mathematics in English for 100 years.

In his 1st term at Trinity, Hardy found himself caught in this system.

He ws to be trained as a racehorse, over a course of mathematical exercises which at 19 he knew to be meaningless.

He was sent to a famous coach, to whom most potential Senior Wranglers went.

This coach knew all the obstacles, all the tricks of the examiners, and was sublimely uninterested in the subject itself.

At this point the young Einstein would have rebelled: he would either have left Cambridge or done no formal work for the next 3 years.

But Hardy was born into the more intensely professional English climate (which was its merits as well as its demerits).

After considering changing his subject to history, he had the sense to find a real mathematician to teach him.

Hardy paid him a tribute in the *Apology*:

> 'My eyes were 1st opened by Prof. Love, who taught me for a few terms and gave me my 1st serious conception of analysis.
>
> But the great debt which I owe to him - he was, after all, primarily an applied mathematician - was his advice to read Jordan's famous *Cours d' Analyse*: and I shall never forget the astonishment with which I read that remarkable work, the 1st inspiration for so many mathematicians of my generation, and learned for the 1st time as I read it what mathematics really meant.

> From that time onwards I was in my way a real mathematician, with sound mathematical ambitions and a genuine passion for mathematics.'

He was 4th Wrangler in 1898.

This faintly irritated him, he used to confess.

He was enough of a natural competitor to feel that, though the race was ridiculous, he ought to have won it.

In 1900 he took Part II of the Tripos, a more respect-worthy examination, and got his right place and his Fellowship.

From that time on, his life was in essence settled.

He knew his purpose, which was to bring rigor into English mathematical analysis.

He did not deviate from the researches which he called 'the 1 great permanent happiness of my life'.

There were no anxieties about what he should do.

Neither he nor anyone else doubted his great talent.

He was elected to the Royal Society at 33.

In many senses, then, he was unusually lucky.

He did not have to think about his career.

From the time he was 23 he had all the leisure that a man could want, and as much money as he needed.

A bachelor don in Trinity in the 1900's was comfortably off.

Hardy was sensible about money, spent it when he felt impelled (sometimes for singular purposes, e.g. 50-mile taxi-rides), and otherwise was not at all unworldly about investments.

He played his games and indulged hi eccentricities.

He was living in some of the best intellectual company in the world - G. E. Moore, Whitehead, Bertrand Russel, Trevelyan, the high Trinity society which was shortly to find its artistic complement in Bloomsbury.

(Hardy himself had links with Bloomsbury, both of personal friendship and of sympathy.)

In a brilliant circle, he was 1 of the most brilliant young men - and, in a quiet way, 1 of the most irrepressible.

I will anticipate now what I shall say later.

His life remained the life of a brilliant young man until he was old: so did his spirit: his games, his interests, kept the lightness of a young don's.

And, like many men who keep a young man's interests into their 60s, his last years were the darker for it.

Much of his life, though, he was happier than most of us.

He had a great many friends, of surprising different kinds.

These friends had to pass some of his private tests: they needed to possess a quality which he called 'spin' (this is a cricket term, and untranslatable: it implies a certain obliquity for irony of approach: of recent public figures, Macmillan and Kennedy would get high marks for spin, Churchill and Eisenhower not).

But he was tolerant, loyal, extremely high-spirited, and in an undemonstrative way found of his friends.

I once was compelled to go and see him in the morning, which was always his set time for mathematical work.

He was sitting at his desk, writing in his beautiful calligraphy.

I murmured some commonplace politeness about hoping that I wasn't disturbing him.

He suddenly dissolved into his mischievous grin.

'As you ought to be able to notice, the answer to that is that you are.

Still, I'm usually glad to see you.'

In the 16 years we knew each other, he didn't say anything more demonstrative than that: except on his deathbed, when he told me that he looked forward to my visits.

I think my experience was shared by most of his close friends.

But he had, scattered through his life, 2 or 3 other relationships, different of kind.

These were intense affections, absorbing, non-physical but exalted.

The one I knew about was for a young man whose nature was as spiritually delicate as his own.

I believe, though I only picked this up from chance remarks, that the same was true of the others.

To many people of my generation, such relationships would seem either unsatisfactory or impossible.

They were neither the one nor the other; and, unless one takes them for granted, one doesn't begin to understand the temperament of men like Hardy (they are rare, but not as rare as white rhinoceroses), nor the Cambridge society of his time.

He didn't get the satisfactions that most of us can't help finding: but he knew himself unusually well, and that didn't make him unhappy.

His inner life was his own, and very rich.

The sadness came at the end.

Apart from his devoted sister, he was left with no one close to him.

With sardonic stoicism he says in the *Apology* - which for all its high spirits is a book of desperate sadness - that when a creative man has lost the power or desire to create - 'It is a pity but in that case he does not matter a great deal anyway, and it would be silly to bother about him.'

That is how he treated his personal life outside mathematics.

Mathematics was his justification.

It was easy to forget this, in the sparkle of his company: just as it was easy in the presence of Einstein's moral passion to forget that to himself his justification was his search for the physical laws.

Neither of those 2 ever forgot it.

This was the core of their lives, from young manhood to death.

Hardly, unlike Einstein, did not make a quick start.

His early papers, between 1990 and 1911, were good enough to get him into the Royal Society and win him an international reputation: but he did not regard them as important.

Again, this wasn't false modesty: it was the judgment of a master who knew to an inch which of his work had value and which hadn't.

In 1911 he began a collaboration with Littlewood which lasted 35 years.

In 1913 he discovered Ramanujan and began another collaboration.

All his major work was done in these 2 partnerships, most of it in the one with Littlewood, the most famous collaboration in the history of mathematics.

There has been nothing like it in any science, or, so far as I know, in any other field of creative activity.

Together they produced nearly a hundred papers, a good many of them 'in the Bradman class'.

Mathematicians not intimate with Hardy in his later years, nor with cricket, keep repeating that is highest term of praise was 'in the Hobbs class'.

It wasn't: very reluctantly, because Hobbs was 1 of his pets, he had to alter the order of merit.

I once had a postcard from him, probably in 1938, saying 'Bradman is a whole class above any batsman who has ever lived: if Archimedes, Newton and Gauss remain in the Hobbs class, I have to admit the possibility of a class above them, which I find difficult to imagine.

They had better be moved from now on into the Bradman class.'

The Hardy-Littlewood researches dominated English pure mathematics, and much of world pure mathematics, for a generation.

It is too early to say, so mathematicians tell me, to what extent they altered the course of mathematical analysis: nor how influential their work will appear in 100 years.

Of its enduring value there is no question.

Theirs was, as I have said, the greatest of all collaborations.

But no one knows how they did it: unless Littlewood tells us, no one will ever know.

I have already given Hardy's judgment that Littlewood was the more powerful mathematician of the 2: Hardy once wrote that he knew of 'no one else who could command such a combination of insight, technique and power.'

Littlewood was and is a more normal man than Hardy, just as interesting and probably more complex.

He never had Hardy's taste for a kind of refined intellectual flamboyance, and so was less in the center of the academic scene.

This led to jokes from European mathematicians, e.g. that Hardy had invented him so as to take the blame in case there turned out anything wrong with 1 of their theorems.

In fact, he is a man of at least as obstinate an individually as Hardy himself.

At 1st glance, neither of them would have seemed the easiest of partners.

It is hard to imagine either of them suggesting the collaboration in the 1st place.

Yet 1 of them must have done.

No one has any evidence how they set about it.

Through their most productive period, they were not at the same university.

Harald Bohr (brother of Niels Bohr, and himself a fine mathematician) is reported as saying that 1 of their principles was this: if one wrote a letter to the other, the recipient was under no obligation to reply to it, or even to read it.

I can't contribute anything.

Hardy talked to me, over a period of many years, on almost every conceivable subject, except the collaboration.

He said, of course, that it had been the major fortune of his creative career: he spoke of Littlewood in the terms I have given: but he never gave a hint of their procedures.

I didn't know enough mathematics to understand their papers, but I picked up some of their language.

If he had let slip anything about their methods, I don't think I should have missed it.

I am fairly certain that the secrecy - quite uncharacteristic of him in matters which to most would seem more intimate - was deliberate.

About his discovery of Ramanujan, he showed no secrecy at all.

It was, he wrote, the 1 romantic incident in his life: anyway, it is an admirable story, and one which showers credit on nearly everyone (with 2 exceptions) in it.

1 morning early in 1913, he found, among the letters on his breakfast table, a large untidy envelope decorated with Indian stamps.

When he opened it, he found sheets of paper by no mean fresh, on which, in a non-English holograph, were line after line of symbols.

Hardy glanced at them without enthusiasm.

He was by this time, at the age of 36, a world famous mathematician: and world famous mathematicians, he had already discovered, are unusually exposed to cranks.

He was accustomed to receiving manuscripts from strangers, proving the prophetic wisdom of the Great Pyramid, the revelations of the Elders of Zion, or the cryptograms that Bacon had inserted in the plays of the so-called Shakespeare.

So Hardy felt, more than anything, bored.

He glanced at the letter, written in halting English, signed by an unknown Indian, asking him to give an opinion of these mathematical discoveries.

The script appeared to consist of theorems, most of them wild or fantastic looking, 1 or 2 already well-known, laid out as though they were original.

There were no proofs by any kind.

Hardy was not only bored, but irritated.

It seemed like a curious kind of fraud.

He put the manuscript aside, and went on with his day's routine.

Since that routine did not vary throughout his life, it is possible to reconstruct it.

1st he read *The Times* over his breakfast.

This happened in January, and if there were any Australian cricket scores, he would start with them, studied with clarity and intense attention.

Maynard Keynes, who began his career as a mathematician and who was a friend of Hardy's, once scolded him: if he had read the stock exchange quotations half an hour each day with the same concentration he brought to the cricket scores, he could not have helped becoming a rich man.

Then, from about 9 to 1, unless he was giving a lecture, he worked at his own mathematics.

4 hours creative work a day is about the limit of a mathematician, he used to say.

Lunch, a light meal, in hall.

After lunch he loped off for a game of real tennis in the university court.

(If it had been summer, he would have walked down to Fenner's to watch cricket.)

In the late afternoon, a stroll back to his rooms.

That particular day, though, while the timetable wasn't altered, internally things were not going according to plan.

At the back of his mind, getting in the way of his complete pleasure in his game, the Indian manuscript nagged away.

Wild theorems.

Theorems such as he had never seen before, nor imagined.

A fraud of genius?

A question was forming itself in his mind.

As it was Hardy's mind, the question was forming itself with epigrammatic clarity: is a fraud of genius more probable than an unknown mathematician of genius?

Clearly the answer was no.

Back in his rooms in Trinity, he had another look at the script.

He sent word to Littlewood (probably by messenger, certainly not by telephone, for which, like all méchanical contrivances including fountain pens, he had a deep distrust) that they must have a discussion after hall.

When the meal was over, there may have been a slight delay.

Hardy liked a glass of wine, but, despite the glorious vistas of 'Alan St. Aubyn' which had fired his youthful imagination, he found he did not really enjoy lingering in the combination-room over port and walnuts.

Littlewood, a good deal more *homme moyen sensuel*, did.

So there may have been a delay.

Anyway, by 9 o'clock or so they were in 1 of Hardy's rooms, with the manuscript stretched out in front of them.

That is an occasion at which one would have liked to be present.

Hardy, with his combination of remorseless clarity and intellectual panache (he was very English, but in argument he showed the characteristics that Latin minds have often assumed to be their own): Littlewood, imaginative, powerful, humorous.

Apparently it did not take them long.

Before midnight they knew, and knew for certain.

The writer of these manuscripts was a man of genius.

That was as much as they could judge, that night.

It was only later that Hardy decided that Ramanujan was, in terms of *natural* mathematical genius, in the class of Gauss and Euler: but that he could not expect, because of the defects of his education, and because he had come on the scene too late in the line of mathematical history, to make a contribution on the same scale.

It all sounds easy, the kind of judgment great mathematicians should have been able to make.

But I mentioned that there were 2 persons who do not come out of the story with credit.

Out of chivalry Hardy concealed this in all that he said or wrote about Ramanujan.

The 2 people concerned have now been dead, however, for many years, and it is time to tell the truth.

It is simple.

Hardy was not the 1st eminent mathematician to be sent the Ramanujan manuscripts.

There had been 2 before him, both English, both of the highest professional standard.

They had each returned the manuscripts without comment.

I don't think history relates what they said, if anything, when Ramanujan became famous.

Anyone who has been sent unsolicited material will have a sneaking sympathy with them.

Anyway, the following day Hardy went into action.

Ramanujan must be brought to England, he decided.

Money was not a major problem.

Trinity has usually been good at supporting unorthodox talent (the college did the same for Kapitsa a few years later).

Once Hardy was determined, no human agency could have stopped Ramanujan, but they needed a certain amount of help from a superhuman one.

Ramanujan turned out to be a poor clerk in Madras, living with his wife on 20 pounds a year.

But he was also a Brahmin, unusually strict about his religious observances, with a mother who was even stricter.

It seemed impossible that he could break the proscriptions and cross the water.

Fortunately his mother had the highest respect for the goddess of Namakkal.

1 morning Ramanujan's mother made a starling announcement.

She had had a dream on the previous night, in which she saw her son seated in a big hall among a group of Europeans, and the goddess of Namakkal had commanded her not to stand in the way of her son fulfilling his life's purpose.

This, say Ramanujan's Indian biographers, was a very agreeable surprise to all concerned.

In 1914 Ramanujan arrived in England.

So far as Hardy could detect (though in this respect I should not trust his insight far) Ramanujan, despite the difficulties of breaking the caste proscriptions, did not believe much in theological doctrine, except for a vague pantheistic benevolence, any more than Hardy did himself.

But he did certainly believe in ritual.

When Trinity put him up in college - within 4 years he became a Fellow - there was no 'Alan St. Aubyn' apolausticity for him at all.

Hardy used to find him ritually changed into his pyjamas, cooking vegetables rather miserably in a frying pan in his own room.

There association was a strangely touching one.

Hardy did not forget that he was in the presence of genius: but genius that was, even in mathematics, almost untrained.

Ramanujan had not been able to enter Madras University because he could not matriculate in English.

According to Hardy's report, he was always amiable and good-natured, but not doubt he sometimes found Hardy's conversation outside mathematics more than a little baffling.

He seems to have listened with a patient smile on his good, friendly, homely face.

Even inside mathematics they had to come to terms with the difference in their education.

Ramanujan was self-taught: he knew nothing of the modern rigor: in a sense he didn't know what a proof was.

In an uncharacteristically sloppy moment, Hardy once wrote that if he had been better educated, he would have been less Ramanujan.

Coming back to his ironic senses, Hardy later corrected himself and said that the statement was nonsense.

If Ramanujan had been better educated, he would have been even more wonderful than he was.

In fact, Hardy was obliged to teach him some formal mathematics as though Ramanujan had been a scholarship candidate at Winchester.

Hardy said that this was the most singular experience of his life: *what did modern mathematics look like to someone who had the deepest insight, but who had literally never heard of most of it?*

Anyway, they produced together 5 papers of the highest class, in which Hardy showed supreme originality of his own (more is known of the details of this collaboration than of the Hardy-Littlewood one).

Generosity and imagination were, for once, rewarded in full.

This is a story of human virtue.

Once people had started behaving well, they went on behaving better.

It is good to remember that England gave Ramanujan such honors as were possible.

The Royal Society elected him a Fellow at the age of 30 (which, even for a mathematician, is very young).

Trinity also elected him a Fellow in the same year.

He was the 1st Indian to be given either of these distinctions.

He was amiably grateful.

But he soon became ill.

It was difficult, in war-time, to move him to a kinder climate.
Hardy used to visit him, as he lay dying in hospital at Putney.
It was on 1 of those visits that there happened the incident of the taxi-cab number.
Hardy had gone out to Putney by taxi, as usual his chosen method of conveyance.
He went into the room where Ramanujan was lying.
Hardy, always inept about introducing a conversation, said, probably without a greeting, and certainly as his 1st remark:

> 'I thought the number of my taxi-cab was 1729.
> It seemed to me rather a dull number.'

To which Ramanujan replied:

> 'No, Hardy! No, Hardy! It is a very interesting number.
> It is the smallest number expressible as the sum of 2 cubes in 2 different ways.'

That is the exchange as Hardy recorded it.
It must be substantially accurate.
He was the most honest of men; and further, no one could possibly have invented it.
Ramanujan died of tuberculosis, back in Madras, 2 years after the war.
As Hardy wrote in the *Apology*, in his roll-call of mathematicians: 'Galois died at 21, Abel at 27, Ramanujan at 33, Riemann at 40... I do not know an instance of a major mathematical advance initiated by a man past 50.'
If it had not been for the Ramanujan collaboration, the 1914-18 war would have been darker for Hardy than it was.
But it was dark enough.
It left a wound which reopened in the 2nd war.
He was a man of radical opinions all his life.
His radicalism, though, was tinged with the turn-of-the-century enlightenment.
To people of my generation, it sometimes seemed to breathe a lighter, more innocent air, than the one we knew.
Like many of his Edwardian intellectual friends, he had a strong feeling for Germany.
Germany had, after all, been the great educating force of the 19th century.
To Eastern Europe, to Russia, to the United States, it was the $\boxed{\textit{German universities which had taught the meaning of research}}$.
Hardy hadn't much use for German philosophy or German literature: his tastes were too classical for that.
But in most respects the German culture, including its social welfare, appeared to him higher than his own.
Unlike Einstein, who had a much more rugged sense of political existence, Hardy did not know much of Wilhelmine Germany at 1st hand.
And, though he was the least vain of people, he would have been less than human if he had not enjoyed being more appreciated in Germany than in his own country.
There is a pleasant anecdote, dating from this period, in which Hilbert, 1 of the greatest of German mathematicians, heard that Hardy lived in a not specially agreeable set of rooms in Trinity (actually in Whewell's Court).
Hilbert promptly wrote in measured terms to the Master, pointing out that Hardy was the best mathematician, not only in Trinity but in England, and should therefore have the best rooms.
So Hardy, like Russell and many of the high Cambridge intelligentsia, did not believe that the war should have been fought.
Further, with his ingrained distrust of English politicians, he though the balance of wrong was on the English side.
He could not find a satisfactory basis for conscientious objection; his intellectual rigor was too strong for that.
In fact, he volunteered for service under the Derby scheme, and was rejected on medical grounds.
But he felt increasingly isolated in Trinity, much of which was vociferously bellicose.
Russell was dismissed from his lectureship, in circumstances of overheated complexity (Hardy was to write the only detailed account of the case a quarter of a century later, in order to comfort himself in another war).
Hardy's close friends were away at the war.
Littlewood was doing ballistics as a 2nd Lieutenant in the Royal Artillery.
Owing to his cheerful indifference he had the distinction of remaining a 2nd Lieutenant through the 4 years of war.
Their collaboration was interfered with, though not entirely stopped.
It was the work of Ramanujan which was Hardy's solace during the bitter college quarrels.
I sometimes thought he was, for once, less than fair to his colleagues.
Some were pretty crazed, as men are in war-time.
But some were long-suffering and tried to keep social manners going.
After all, it was a triumph of academic uprightness that they should have elected his protégé Ramanujan, at a time when Hardy was only just on speaking terms with some of the electors, and not at all with others.
Still, he was harshly unhappy.
As soon as he could, he left Cambridge.

He was offered a chair at Oxford in 1919: and immediately walked into the happiest time of his life.

He had already done great work with Ramanujan and Littlewood rose to its full power.

Hardy was, in Newton's phrase, 'in the prime of his age for invention', and this came in his early 40s, unusually late for a mathematician.

Coming so late, this creative surge gave him the feeling, more important to him than to most men, of timeless youth.

He was living the young man's life which was 1st nature to him.

He played more real tennis, and got steadily better at it (real tennis was an expensive game and took a largish slice out of a professional income).

He made a good many visits to American universities, and loved the country.

He was 1 of the few Englishmen of his time who was fond, to an extent approximately equal, of the United States and the Soviet Union.

He was certainly the only Englishman of his or any other time who wrote a serious suggestion to the Baseball Commissioners, proposing a technical emendation to 1 of the rules.

The 20s, for him and for most liberals of his generation, was a false dawn.

He thought the misery of the war was swept away into the past.

He was at home in New College as he had never been in Cambridge.

The warm domestic conversational Oxford climate was good for him.

It was there, in a college at that time small and intimate, that he perfected his own style of conversation.

There was company eager to listen to him after hall.

They could take his eccentricities.

He was not only a great and good man, they realized, but an entertaining one.

If he wanted to play conversational games, or real (though bizarre) games on the cricket field, they were ready to act as foils.

In a casual and human fashion, they made a fuss of him.

He had been admired and esteemed before, but not made a fuss of in that fashion.

No one seemed to care - it was a gossipy college joke - that he had a large photograph of Lenin in his rooms.

Hardy's radicalism was somewhat unorganized, but it was real.

He had been born, as I have explained, into a professional family: almost all his life was spent among the *haute bourgeoisie*: but in fact he behaved much more like an aristocrat, or more exactly like 1 of the romantic projections of an aristocrat.

Some of this attitude, perhaps, he had picked up from his friend Bertrand Russell.

But most of it was innate.

Underneath his shyness, he just didn't give a damn.

He got on easily, without any patronage, with the poor, the unlucky and diffident, those who were handicapped by race (it was a symbolic stroke of fate that he discovered Ramanujan).

He preferred them to the people whom he called the *large bottomed*: the description was more psychological than anatomical, though there was a famous 19th-century Trinity aphorism by Adam Sedgwick: 'No one ever made a success in this world without a large bottom.'

To Hardy the large bottomed were the confident, booming, imperialist bourgeois English.

The designation included most bishops, headmasters, judges, and all politicians, with the single exception of Lloyd George.

Just to show his allegiances, he accepted 1 publice office.

For 2 years (1924-26) he was President of the Association of Scientific Workers.

He said sarcastically he was an odd choice, being 'the most unpractical member of the most unpractical profession in the world'.

But in important things he was not so unpractical.

He was deliberately standing up to be counted.

When, much later, I came to work with Frank Cousins, it gave me a certain quiet pleasure to reflect that I had had exactly 2 friends who had held office in the Trade Union movement, him and G. H. Hardy.

That late, not quite Indian, summer in Oxford in the 20s was so happy that people wondered that he ever returned to Cambridge.

Which he did in 1931.

I think there were 2 reasons.

1st and most decisive, he was a great professional.

Cambridge was still the center of English mathematics, and the senior mathematical chair there was the correct place for a professional.

2ndly, and rather oddly, he was thinking about his old age.

Oxford colleges, in many ways so human and warm, are ruthless with the old: if he remained at New College he would be turned out of his rooms as soon as he retired, under the age limit, from his professorship.

Whereas, if he returned to Trinity, he could stay in college until he died.

That is in effect what he managed to do.

When he came back to Cambridge - which was the time that I began to know him - he was in the afterglow of his great period.

He was still happy.

He was still creative, not so much as in the 20s, but enough to make him feel that the power was still there.

He was as spirited as he had been at New College.

So we had the luck to see him very neatly at his best.

In the winters, after we had become friendly, we gave each other dinner in our respective colleges once a fortnight.

When summer came, it was taken for granted that we should meet at th cricket ground.

Except on special occasions he still did mathematics in the morning, and did not arrive at Fenner's until after lunch.

He used to walk around the cinderpath with a long, loping, clumping-footed stride (he was a slight spare man, physically active even in his late 50s, still playing real tennis) head down, hair, tie, sweaters, papers all flowing, a figure that caught everyone's eyes.

'There goes a Greek poet, I'll be bound,' once said some cheerful farmer as Hardy passed the score-board.

He made for his favorite place, opposite the pavilion, where he could catch each ray of sun - he was obsessively heliotropic.

In order to deceive the sun into shining, he brought with him, even on a fine May afternoon, what he called his 'anti-God battery'.

This consisted of 3 or 4 sweaters, an umbrella belonging to his sister, and a large envelope containing mathematical manuscripts, e.g. a Ph.D dissertation, a paper which he was refereeing for the Royal Society, or some tripos answers.

He would explain to an acquaintance that God, believing that Hardy expected the weather to change and give him a chance to work, counter-suggestibly arranged that the sky should remain cloudless.

There he sat.

To complete his pleasure in a long afternoon watching cricket, he liked the sun to be shining and a companion to join in the fun.

Technique, tactics, formal beauty - those were the deepest attractions of the game for him.

I won't try to explain them: they are incommunicable unless one knows the language: just as some of Hardy's classical aphorisms are inexplicable unless one knows either the language of cricket or of the theory of numbers, and preferably both.

Fortunately for a good many of our friends, he also had a relish for the human comedy.

He would have been the 1st to disclaim that he had any special psychological insight.

But he was the most intelligent of men, he had lived with his eyes open and read a lot, and he had obtained a good generalized sense of human nature - robust, indulgent, satirical, and utterly free from moral vanity.

He was spiritually candid as few men are (I doubt if anyone could be more candid), and he had a mocking horror of pretentiousness, self-righteous indignation, and the whole stately pantechnicon of the hypocritical virtues.

Now cricket, the most beautiful of games, is also the most hypocritical.

It is supposed to be the ultimate expression of the team spirit.

One ought to prefer to make 0 and see one's side win, than make 100 and see it lose (one very great player, like Hardy a man of innocent candor, once remarked mildly that he never managed to feel so).

This particular ethos inspired Hardy's sense of the ridiculous.

In reply he used to expound a counterbalancing series of maxims.

Example 0.1. *'Cricket is the only game where you are playing against 11 of the other side and 10 of your own.'*

'If you are nervous when you go in 1st, nothing restores your confidence so much as seeing the other man get out.'

If his listeners were lucky, they would hear other remarks, not relevant to cricket, as sharp-edged in conversation as in his writing.

In the *Apology* there are some typical specimens: here are a few more.

> 'It is never worth a 1st class man's time to express a majority opinion.
> By definition, there are plenty of others to do that.'
>
> 'When I was an undergraduate one might, if one were sufficiently unorthodox, suggest that Tolstoi came within touching distance of George Meredith as a novelist: but, of course, no one else possibly could.'
>
> (This was said about the intoxications of fashion: it is worth remembering that he had lived in 1 of the most brilliant of all Cambridge generations.)
>
> 'For any serious purpose, intelligence is a very minor gift.'
>
> 'Young men ought to be conceited: but they oughtn't to be imbecile.'
>
> (Said after someone had tried to persuade him that *Finnegans Wake* was the final literary masterpiece.)
>
> 'Sometimes one has to say difficult things, but one ought to say them as simply as one knows how.'

Occasionally, as he watched cricket, his ball-by-ball interest flagged.

Then he demanded that we should pick teams: teams of humbugs, club-men, bogus poets, bores, teams whose names began with HA (numbers 1 and 2 Hadrican and Hannibal), SN, all-time teams of Trinity, Christ's, and so on.

In these exercises I was at a disadvantage: let anyone try to produce a team of world-figures whose names start with SN.

The Trinity team is overwhelmingly strong (Clerk Maxwell, Byron, Thackeray, Tennyson aren't certain of places): while Christ's, beginning strongly with Milton and Darwin, has nothing much to show from number 3 down.

Or he had another favorite entertainment.

'Mark that man we met last night', he said, and someone had to be marked out of 100 in each of the categories Hardy had long since invented and defined.

Stark, Bleak ('a stark man is not necessarily bleak: but all bleak men without exception want to be considered stark'), Dim, Old Brandy, Spin, and some others.

Stark, Bleak and Dim are self-explanatory (the Duke of Wellington would get a flat 100 for Stark and Bleak, and 0 for Dim).

Old brandy was derived from a mythical character who said that he never drank anything but old brandy.

Hence, by extrapolation, Old Brandy came to mean a taste that was eccentric, esoteric, but just within the bounds of reason.

As a character (and in Hardy's view, though not mine, as a writer also) Proust got high marks for Old Brandy: so did F. A. Lindemann (later Lord Cherwell).

The summer days passed.

After 1 of the short Cambridge seasons, there was the University match.

Arranging to meet him in London was not always simple, for, as I mentioned before, he had a morbid suspicion of mechanical gadgets (he never used a watch), in particular of the telephone.

In his rooms in Trinity or his flat in St George's Square, he used to say, in a disapproving and slightly sinister tone: 'If you *fancy yourself* at the telephone, there is one in the next room.'

Once in an emergency he had to ring me up: angrily his voice came at me:

> 'I shan't hear a word you say, so when I'm finished I shall immediately put the receiver down.
> It's important you should come round between 9 and 10 tonight.'

Clonk.

Yet, punctually, he arrived at the University match.

There he was at his most sparkling, year after year.

Surrounded by friends, men and women, he was quite released from shyness.

He was the center of all our attention, which he didn't dislike.

Sometimes one could hear the party's laughter from a quarter of the way round the ground.

In those last of his happy years, everything he did was light with grace, order, a sense of style.

Cricket is a game of grace and order, which is why he found formal beauty in it.

His mathematics, so I am told, had these same aesthetic qualities, right up to his last creative work.

I have given the impression, I fancy, that in private he was a conversational performer.

To an extent, that was true: but he was also, on what he would have called 'non-trivial' occasions (meaning occasions important to either participant) a serious and concentrated listener.

Of other eminent persons whom, by various chances, I knew at the same period, Wells was, on the whole, a worse listener than one expected: Rutherford distinctly better: Lloyd George 1 of the best listeners of all time.

Hardy didn't suck impressions and knowledge out of others' words, as Lloyd George did, but his mind was at one's disposal.

When, years before I wrote it, he heard of the concept of *The Masters*, he cross-examined me, so that I did most of the talking.

He produced some good ideas.

I wish he had been able to read the book, which I think that he might have liked.

Anyway, in that hope I dedicated it to his memory.

In the note at the end of the *Apology* he refers to other discussions.

1 of these was long-drawn-out, and sometimes, on both sides, angry.

On the 2nd world war we each had passionate but, as I shall have to say a little later, different opinions.

I didn't shift his by 1 inch.

Nevertheless, though we were separated by a gulf of emotion, on the plane of reason he recognized what I was saying.

That was true in any argument I had with him.

Through the 30s he lived his own version of a young man's life.

Then suddenly it broke.

In 1939 he had a coronary thrombosis.

He recovered, but real tennis, squash, the physical activities he loved, were over for good.

The war darkened him still further, just as the 1st war had.

To him they were connected pieces of lunacy, we were all at fault, he couldn't identify himself with the war - once it was clear the country would survive - any more than he had done in 1914.

1 of his closest friends died tragically.

And - I think there is no doubt these griefs were inter-connected - his creative powers as a mathematician at last, in his 60s, left him.

That is why *A Mathematician's Apology* is, if read with the textual attention it deserves, a book of haunting sadness.

Yes, it is witty and sharp with intellectual high spirits: yes, the crystalline clarity and candor are still there: yes, it is the testament of a creative artist.

But it is also, in an understated stoical fashion, a passionate lament for creative powers that used to be and that will never come again.

I know nothing like it in the language: partly because most people with the literary gift to express such a lament don't come to feel it: it is very rare for a writer to realize, with the finality of truth, that he is absolutely finished.

Seeing him in those years, I couldn't help thinking of the price he was paying for his young man's life.

It was like seeing a great athlete, for years in the pride of his youth and skill, so much younger and more joyful than the rest of us, *suddenly have to accept that the gift has gone.*

It is common to meet great athletes who have gone, as they call it, over the hill: fairly quickly the feet get heavier (often the eyes last longer), the strokes won't come off, Wimbledon is a place to be dreaded, the crowds go to watch someone else.

That is the point at which a good many athletes take to drink.

Hardy didn't take to drink: but he took to something like despair.

He recovered enough physically to have 10 minutes batting at the nets, or to play his pleasing elaboration (with a complicated set of bisques) of Trinity bowls.

But it was often hard to rouse his interest - 3 or 4 years before his interest in everything was so sparkling as sometimes to tire us all out.

'No one should ever be bored', had been 1 of his axioms.

'One can be horrified, or disgusted, but one can't be bored.'

Yet now he was often just that, plain bored.

It was for that reason that some of his friends, including me, encouraged him to write the story of Bertrand Russell and Trinity in the 1914-18 war.

People who didn't know how depressed Hardy was thought the whole episode was now long over and ought not to be resurrected.

The truth was, it enlivened him to have any kind of purpose.

The book was privately circulated.

It has never been obtainable by the public, which is a pity, for it is a small-scale addition to academic history.

I used such persuasion as I had to get him to write another book, which he had in happier days promised me to do.

It was to be called *A Day at the Oval* and was to consist of himself watching cricket for a whole day, spreading himself in disquisitions on the game, human nature, his reminiscences, life in general.

It would have been an eccentric minor classic: but it was never written.

I wasn't much help to him in those last years.

I was deeply involved in war-time Whitehall, I was preoccupied and often tired, it was an effort to get to Cambridge.

But I ought to have made the effort more often than I did.

I have to admit, with remorse, that there was, not exactly a chill, but a gap in sympathy between us.

He lent me his flat in Pimlico - a dark and seedy flat with the St George's Square gardens outside and what he called an 'old brandy' attractiveness - for the whole of war.

But he didn't like me being so totally committed.

People he approved of oughtn't to give themselves whole-heartedly to military functions.

He never asked me about my work.

He didn't want to talk about the war.

While I, for my part, was impatient and didn't show anything like enough consideration.

After all, I thought, I wasn't doing this job for fun: as I had to do it, I might as well extract the maximum interest.

But that is no excuse.

At the end of the war I did not return to Cambridge.

I visited him several times in 1946.

His depression had not lifted, he was physically failing, short of breath after a few yards walk.

The long cheerful stroll across Parker's Piece, after the close of play, was gone for ever: I had to take him home to Trinity in a taxi.

He was glad that I had gone back to writing books: the creative life was the only one for a serious man.

As for himself, he wished that he could live the creative life again, no better than it had been before: his own life was over.

I am not quoting his exact words.

This was so unlike him that I wanted to forget, and I tried, by a kind of irony, to smear over what had just been said.

So that I have never remembered precisely.

I attempted to dismiss it to myself as a rhetorical flourish.

In the early summer of 1947 I was sitting at breakfast when the telephone rang.

It was Hardy's sister: he was seriously ill, would I come up to Cambridge at once, would I call at Trinity 1st?

At the time I didn't grasp the meaning of the 2nd request.

But I obeyed it, and in the porter's lodge at Trinity that morning found a note from her: I was to go to Donald Robertson's rooms, he would be waiting for me.

Donald Robertson was the Professor of Greek, and an intimate friend of Hardy's: he was another member of the same high, liberal, graceful Edwardian Cambridge.

Incidentally, he was 1 of the few people who called Hardy by his Christian name.

He greeted me quietly.

Outside the windows of his room it was a calm and sunny morning.

He said:

'You ought to know that Harold has tried to kill himself.'

Yes, he was out of danger: he was for the time being all right, if that was the phrase to use.

But Donald was, in a less pointed fashion, as direct as Hardy himself.

It was a pity the attempt had failed.

Hardy's health had got worse: he could not in any case live long: even walking from his rooms to hall had become a strain.

He had made a completely deliberate choice.

Life on those terms he would not endure: there was nothing in it.

He had collected enough barbiturates: he had tried to do a thorough job, and had taken too many.

I was fond of Donald Robertson, but I had met him only at parties and at Trinity high table.

This was the 1st occasion on which we had talked intimately.

He said, with gentle firmness, that I ought to come up to see Hardy as often as I could: it would be hard to take, but it was an obligation: probably it would not be for long.

We were both wretched.

I said goodbye, and never saw him again.

In the Evelyn nursing home, Hardy was lying in bed.

As a touch of farce, he had a black eye.

Vomiting from the drugs, he had hit his head on the lavatory basin.

He was self-mocking.

He had made a mess of it.

Had anyone ever made a bigger mess?

I had to enter into the sarcastic game.

I had never felt less like sarcasm, but I had to play.

I talked about other distinguished failures at bringing off suicide.

What about German generals in the last war?

Beck, Stulpnagel, they had been remarkably incompetent at it.

It was bizarre to hear myself saying these things.

Curiously enough, it seemed to cheer him up.

After that, I went to Cambridge at least once a week.

I dreaded each visit, but early on he said that he looked forward to seeing me.

He talked a little, nearly every time I saw him, about death.

He wanted it: he didn't fear it: what was there to fear in nothingness?

His hard intellectual stoicism had come back.

He would not try to kill himself again.

He wasn't good at it.

He was prepared to wait.

With an inconsistency which might have pained him - for he, like most of his circle, believed in the rational to an extent that I thought irrational - he showed an intense hypochondriac curiosity about his own symptoms.

Constantly he was studying the oedema of his ankles: was it greater or less that day?

Mostly, though - about 55 minutes in each hour I was with him - I had to talk cricket.

It was his only solace.

I had to pretend a devotion to the game which I no longer felt, which in fact had been lukewarm in the 30s except for the pleasure of his company.

Now I had to study the cricket scores as intently as when I was a schoolboy.

He couldn't read for himself, but he would have known if I was bluffing.

Sometimes, for a few minutes, his old vivacity would light up.

But if I couldn't think of another question or piece of news, he would lie there, in the kind of dark loneliness that comes to some people before they die.

Once or twice I tried to rouse him.

Wouldn't it be worth while, even if it was a risk, to go and see 1 more cricket match together?

I was now better off than I used to be, I said.

I was prepared to stand him a taxi, his old familiar means of transport, to any cricket ground he liked to name.

At that he brightened.

He said that I might have a dead man on my hands.

I replied that I was ready to cope.

I thought that he might come: he knew, I knew, that his death could only be a matter of months: I wanted to see him have 1 afternoon of something like gaiety.

The next time I visited him he shook his head in sadness and anger.

No, he couldn't even try: there was no point in trying.

It was hard enough for me to have to talk cricket.

It was harder for his sister, a charming intelligent woman who had never married and who had spent much of her life looking after him.

With a humorous skill not unlike his own old form, she collected every scrap of cricket news she could find, though she had never learned anything about the game.

Once or twice the sarcastic love of the human comedy came bursting out.

2 or 3 weeks before his death, he heard from the Royal Society that he was to be given their highest honor, the Copley Medal.

He gave his Mephistophelian grin, the 1st time I had seen it in full splendor in all those months.

> 'Now I know that I must be pretty near the end.
> When people hurry up to give you honorific things there is exactly 1 conclusion to be drawn.'

After I heard that, I think I visited him twice.

The last time was 4 or 5 days before he died.

There was an Indian test team playing in Australia, and we talked about them.

It was in that same week that he told his sister: 'If I knew that I was going to die today, I think I should still want to hear the cricket scores.'

He managed something very similar.

Each evening that week before she left him, she read a chapter from a history of Cambridge university cricket.

1 such chapter contained the last words he heard, for he died suddenly, in the early morning."

C. P. Snow

Preface

I am indebted for many valuable criticisms to Prof. C. D. Broad and Dr. C. P. Snow, each of whom read my original manuscript.

I have incorporated the substance of nearly all of their suggestions in my text, and have so removed a good many crudities and obscurities.

In 1 case I have dealt with them differently.

My §28 is based on a short article which I contributed to *Eureka* (the journal of the Cambridge Archimedean Society) early in the year, and I found it impossible to remodel what I had written so recently and with so much care.

Also, if I had tried to meet such important criticisms seriously, I should have had to expand this section so much as to destroy the whole balance of my essay.

I have therefore left it unaltered, but have added a short statement of the chief points made by my critics in a note at the end.

G. H. H
Jul 18, 1940

1

It is a melancholy experience for a professional mathematician to find himself writing about mathematics.

The function of a mathematician is to do something, to prove new theorems, to add to mathematics, and not to talk about what he or other mathematicians have done.

Statesmen despise publicists, painters despise art-critics, and physiologists, physicists, or mathematicians have usually similar feelings; there is no scorn more profound, or on the whole more justifiable, than that of the men who make for the men who explain.

Exposition, criticism, appreciation, is work for 2nd-rate minds.

I can remember arguing this point once in 1 of the few serious conversations that I ever had with Housman.

Housman, in his Leslie Stephen lecture *The Name & Nature of Poetry*, had denied very emphatically that he was a 'critic'; but he had denied it in what seemed to me a singularly perverse way, and had expressed an admiration for literary criticism which startled and scandalized me.

He had begun with a quotation from his inaugural lecture, delivered 22 years before -

> Whether the faculty of literary criticism is the best gift that Heaven has in its treasuries, I cannot say; but Heaven seems to think so, for assuredly it is the gift most charily bestowed.
>
> Orators and poets. . ., if rare in comparison with blackberries, are commoner than returns of Halley's comet: literary critics are less common. . .

And he had continued -

> In these 22 years I have improved in some respects and deteriorated in others, but I have not so much improved as to become a literary critic, nor so much deteriorated as to fancy that I have become one.

It had seemed to me deplorable that a great scholar and a fine poet should write like this, and, finding myself next to him in Hall a few weeks later, I plunged in and said so.

Did he really mean what he had said to be taken very seriously?

Would the life of the best of critics really have seemed to him comparable with that of a scholar and a poet?

We argued these questions all through dinner, and I think that finally he agreed with me.

I must not seem to claim a dialectical triumph over a man who can no longer contradict me; but 'Perhaps not entirely' was, in the end, his reply to the 1st question, and 'Probably no' to the 2nd.

There may have been some doubt about Housman's feelings, and I do not wish to claim him as on my side; but there is no doubt at all about the feelings of men of science, and I share them fully.

If then I find myself writing, not mathematics but 'about' mathematics, it is a confession of weakness, for which I may rightly be scorned or pitied by younger and more vigorous mathematicians.

I write about mathematics because, like any other mathematician who has passed 60, I have no longer the *freshness of mind*, the *energy*, or the *patience to carry on effectively* with my *proper job*.

2

I propose to put forward an apology for mathematics; and I may be told that it needs none, since there are now few studies more generally recognized, for good reasons or bad, as profitable and praiseworthy.

This may be true; indeed it is probable, since the sensational triumphs of Einstein, that stellar astronomy and atomic physics are the only sciences which stand higher in popular estimation.

A mathematician need no now consider himself on the defensive.

He does not have to meet the sort of opposition described by Bradley in the admirable defense of metaphysics which forms the introduction to *Appearance and Reality*.

A metaphysician, says Bradley, will be told that 'metaphysical knowledge is wholly impossible', or that 'even if possible to a certain degree, it is practically no knowledge worth the name'.

'The same problems,' he will hear, 'the same disputes, the same sheer failure.

Why not abandon it and come out?

Is there nothing else more worth your labor?'

There is no one so stupid as to use this sort of language about mathematics.

The mass of mathematical truth is obvious and imposing; its practical applications, the bridges and steam-engines and dynamos, obtrude themselves on the dullest imagination.

The public does not need to be convinced that there is something in mathematics.

All this is in its way very comforting to mathematicians, but it is hardly possible for a genuine mathematician to be content with it.

Any genuine mathematician must feel that it is not on these crude achievements that the real case for mathematics rests, that the popular reputation of mathematics is based largely on ignorance and confusion, and that there is room for a more rational defense.

At any rate, I am dispose to try to make one.

It should be a simpler task than Bradley's difficult apology.

I shall ask, then, *why is it really worth while to make a serious study of mathematics?*

What is the proper justification of a mathematician's life?

And my answers will be, for the most part, such as are to be expected from a mathematician: I think that it is worth while, that there is ample justification.

But I should say at once that my defense of mathematics will be a defense of myself, and that my apology is bound to be to some extent egotistical.

I should not think it worth while to apologize for my subject if I regarded myself as 1 of its failures.

Some egotism of this sort is inevitable, and I do not feel that it really needs justification.

Good work is not done by 'humble' men.

It is 1 of the 1st duties of a professor, e.g., in any subject, to exaggerate a little both the importance of his subject and his own importance in it.

A man who is always asking '*Is what I do worth while?*' and '*Am I the right person to do it?*' will always be ineffective himself and a discouragement to others.

He must shut his eyes a little and think a little more of his subject and himself than they deserve.

This is not too difficult: it is harder not to make his subject and himself ridiculous by shutting his eyes too tightly.

3

A man who sets out to justify his existence and his activities has to distinguish 2 different questions.

The 1st is whether the work which he does is worth doing; and the 2nd is why he does it, whatever its value may be.

The 1st question is often very difficult, and the answer very discouraging, but most people will find the 2nd easy enough even then.

Their answers, if they are honest, will usually take 1 or other of 2 forms; and the 2nd form is merely a humbler variation of the 1st, which is the only answer which we need consider seriously.

> (I) 'I do what I do because it is the one and only thing that I can do at all well.
>
> I am a lawyer, or a stockbroker, or a professional cricketer, because I have some real talent for that particular job.
>
> I am a lawyer because I have a fluent tongue, and am interested in legal subtleties; I am a stockbroker because my judgment of the markets is quick and sound; I am a professional cricketer because I can bat unusually well.
>
> I agree that it might be better to be a poet or a mathematician, but unfortunately I have no talent for such pursuits.'

I am not suggesting that this is a defense which can be made by most people, since most people can do nothing at all well.

But it is impregnable when it can be made without absurdity, as it can by a substantial minority: perhaps 5 or even 10% of men can do something rather well.

It is a tiny minority who can do anything *really* well, and the number of men who can do 2 things well is negligible.

If a man has any genuine talent, he should be ready to make almost any sacrifice in order to cultivate it to the full.

This view was endorsed by Dr. Johnson -

> When I told him that I had been to see [his namesake] Johnson ride upon 3 horses, he said 'Such a man, sir, should be encouraged, for his performances show the extent of the human powers...' -

and similarly he would have applauded mountain climbers, channel swimmers, and blindfold chess-players.

For my own part, I am entirely in sympathy with all such attempts at remarkable achievement.

I feel some sympathy even with conjurors and ventriloquists; and when Alekhine and Bradman set out to beat records, I am quite bitterly disappointed if they fail.

And here both Dr. Johnson and I find ourselves in agreement with the public.

As W. J. Turner has said so truly, it is only the 'highbrows' (in the unpleasant sense) who do not admire the 'real swells'.

We have of course to take account of the differences in value between different activities.

I would rather be a novelist or a painter than a statesman of similar rank; and there are many roads to fame which most of us would reject as actively pernicious.

Yet it is seldom that such differences of value will turn the scale in a man's choice of a career, which will almost always be dictated by the limitations of his natural abilities.

Poetry is more valuable than cricket, but Bradman would be a fool if he sacrificed his cricket in order to write 2nd-rate minor poetry (and I suppose that it is unlikely that he could do better).

If the cricket were a little less supreme, and the poetry better, then the choice might be more difficult: I do not know whether I would rather have been Victor Trumper or Rupert Brooke.

It is fortunate that such dilemmas occur so seldom.

I may add that they are particularly unlikely to present themselves to a mathematician.

It is usual to exaggerate rather grossly the differences between the mental processes of mathematicians and other people, but it is undeniable that a gift for mathematics is 1 of the most specialized talents, and that mathematicians as a class are not particularly distinguished for general ability or versatility.

If a man is in any sense a real mathematician, then it is a hundred to 1 that his mathematics will be far better than anything else he can do, and that he would be silly if he surrendered any decent opportunity of exercising his 1 talent in order to do undistinguished work in other fields.

Such a sacrifice could be justified only by economic necessity or age.

4

I had better say something here about this question of age, since it is particularly important for mathematicians.

No mathematician should ever allow himself to forget that mathematics, more than any other art of science, is a young man's game.

To take a simple illustration at a comparatively humble level, the average age of election to the Royal Society is lowest in mathematics.

We can naturally find much more striking illustrations.

We may consider, e.g., the career of a man who was certainly 1 of the world's 3 greatest mathematicians.

Newton gave up mathematics at 50, and had lost his enthusiasm long before; he had recognized no doubt by the time that he was 40 that his great creative days were over.

His greatest ideas of all, fluxions and the law of gravitation, came to him about 1666, when he was 24 - 'in those days I was in the prime of my age for invention, and minded mathematics and philosophy more than at any time since'.

He made big discoveries until he was nearly 40 (the 'elliptic orbit' at 37), but after that he did little but polish and perfect.

Galois died at 21, Abel at 27, Ramanujan at 33, Riemann at 40.

There have been men who have done great work a good deal later; Gauss's great memoir on differential geometry was published when he was 50 (though he had had the fundamental ideas 10 years before).

I do not know an instance of a major mathematical advance initiated by a man past 50.

If a man of mature age loses interest in and abandons mathematics, the loss is not likely to be very serious either for mathematics or for himself.

On the other hand the gain is no more likely to be substantial; the later records of mathematicians who have left mathematics are not particularly encouraging.

Newton made a quite competent Master of the Mint (when he was not quarreling with anybody).

Painlevé was a not very successful Premier of France.

Laplace's political career was highly discreditable, but he is hardly a fair instance, sine he was dishonest rather than incompetent, and never really 'gave up' mathematics.

It is very hard to find an instance of a 1st-rate mathematician who has abandoned mathematics an attained 1st-rate distinction in any other field.[1]

There may have been young men who would have been 1st-rate mathematicians if they had stuck to mathematics, but I have never heard of a really plausible example.

And all this is fully borne out by my own very limited experience.

Every young mathematician of real talent whom I have known has been faithful to mathematics, and not from lack of ambition but from abundance of it; they have all recognized that there, if anywhere, lay the road to a life of any distinction.

[1] Pascal seems the best.